# Data Mining 1 (Istraživanje podataka 1) – Comprehensive Summary (Atlas)

Generated from course script (PDF) – English-first outline

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## Contents

# 1 Introduction (Uvod)

- **Data mining / KDD:** non-trivial extraction of *implicit*, *previously unknown*, and *potentially useful* information from data.
- **Why it matters:** data volume grows fast; traditional analysis struggles with scale + spatio-temporal structure; much *raw data* never gets analyzed.
- **Related areas:** statistics, AI, machine learning, pattern recognition, DB technologies, parallel/distributed computing; also *Big Data*, *Predictive Analytics*, *Data Science*.
- **Typical pipeline:** storage → transformation → cleaning → re-storage/use → mining/analysis; key challenges include heterogeneity, scalability, dimensionality, data quality (missing/irrelevant/noisy), interpretation/visualization, privacy/security.

## 1.1 Main building blocks / tasks

- **Data (Podaci):** usually arranged as a matrix (records × attributes); preprocessing is critical.
- **Association rules (Pravila pridruživanja):** discover relations like $A \to B$.
- **Classification (Klasifikacija, supervised):** learn a mapping from attributes $X$ to a target $y$ (categorical for classification, numeric for regression).
- **Clustering (Klasterovanje, unsupervised):** group instances by similarity.
- **Anomaly detection (Otkrivanje anomalija):** find instances that strongly deviate from the rest (outliers).
- **Analysis & visualization:** interpret patterns and results.

# 2 Data (Podaci)

## 2.1 Objects, attributes, and attribute types

- **Dataset:** objects (records/instances) described by attributes (features).
- **Attribute value types (by allowed operations):**
  - **Nominal (Imenski):** equality/inequality ($=, \neq$). Examples: ID, ZIP, eye color.
  - **Ordinal (Redni):** ordered categories ($<, \leq, >, \geq$). Examples: grades/ranks.
  - **Interval (Intervalni):** meaningful differences ($+, -$), no true zero. Examples: dates, temperature.
  - **Ratio (Razmerni):** meaningful ratios ($\cdot, /$). Examples: length, mass, counts.
- **Discrete vs continuous:** discrete values (incl. binary); continuous values (real-valued).
- **Asymmetric binary:** only presence (1) matters; 0–0 matches often uninformative.

## 2.2 Types of datasets

- **Independent/tabular data:** multi-dimensional or text; often sparse (many zeros).
- **Dependent data:** implicit dependencies (e.g., sensors), explicit dependencies (graphs/networks).
- **Ordered data:** time series (ECG, temperature), sequences (DNA/RNA), spatial and spatio-temporal data.
- **Graph (network) data:** nodes are objects, edges represent relations; can be directed/weighted.

# 3 Similarity and Dissimilarity Measures (Mere sličnosti)

## 3.1 Basics

- **Similarity** $s$ typically in $[0, 1]$; **distance/dissimilarity** $d$ typically in $[0, \infty)$.
- **Metric axioms:** non-negativity and identity, symmetry, triangle inequality.
- **Ultrametric:** stronger form: $d(p, r) \leq \max\{d(p, q), d(q, r)\}$.

### 3.2 Distances for continuous variables

- **Minkowski ($L_p$):**

$$d(X,Y) = \left( \sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}$$

- Special cases:
  - **Manhattan** ($p = 1$): $d(X,Y) = \sum_i |x_i - y_i|$
  - **Euclidean** ($p = 2$): $d(X,Y) = \sqrt{\sum_i (x_i - y_i)^2}$
  - **Chebyshev** ($p \to \infty$): $d(X,Y) = \max_i |x_i - y_i|$
  - **Hamming** (for symbolic/binary vectors): $d(X,Y) = \sum_i \mathbf{1}[x_i \neq y_i]$
- **Weighted Minkowski:**

$$d(X,Y) = \left( \sum_{i=1}^{n} a_i |x_i - y_i|^p \right)^{1/p}$$

- **Mahalanobis:**

$$d(X,Y) = \sqrt{(X-Y)\,\Sigma^{-1}\,(X-Y)^T}$$

  Useful with correlated attributes, different scales, (approximately) normal distributions; expensive to compute.

### 3.3 Binary / categorical proximity

Let $M_{01}, M_{10}, M_{00}, M_{11}$ count matches/mismatches between two binary vectors.

- **Simple Matching Coefficient (SMC):**

$$SMC = \frac{M_{11} + M_{00}}{M_{00} + M_{01} + M_{10} + M_{11}}$$

- **Jaccard (asymmetric):**

$$J = \frac{M_{11}}{M_{01} + M_{10} + M_{11}}$$

- **Tanimoto / extended Jaccard:**

$$T(X,Y) = \frac{X \cdot Y}{\|X\|^2 + \|Y\|^2 - X \cdot Y}$$

- **Cosine similarity:**

$$\cos(X,Y) = \frac{X \cdot Y}{\|X\|\,\|Y\|}$$

### 3.4 Correlation

- **Pearson correlation:**

$$\rho_{X,Y} = \frac{\mathrm{cov}(X,Y)}{\sigma_X \sigma_Y}, \qquad \mathrm{cov}(X,Y) = \frac{\sum_{k=1}^{n} (x_k - \bar{X})(y_k - \bar{Y})}{n-1}$$

- Interpretation: $\rho = 1$ (perfect positive linear), $\rho = -1$ (perfect negative linear).

### 3.5 Document / string similarity

- **TF–IDF-style weighting:**

$$idf_i = \log \frac{n}{n_i}, \qquad h(x_i) = f(x_i)\, idf_i$$

  with typical dampening $f(x) = \sqrt{x}$ or $f(x) = \log x$.

- **Cosine for weighted document vectors:**

$$\cos(X,Y) = \frac{\sum_i h(x_i)h(y_i)}{\sqrt{\sum_i h(x_i)^2}\,\sqrt{\sum_i h(y_i)^2}}$$

- **Edit distance** (costs: delete $C_b$, insert $C_u$, replace $C_z$):

$$Edit(i,j) = \min \begin{cases} Edit(i-1,j) + C_b \\ Edit(i,j-1) + C_u \\ Edit(i-1,j-1) + I_{ij}C_z \end{cases}$$

- **LCSS (Longest Common Subsequence):**

$$LCSS(i,j) = \max \begin{cases} LCSS(i-1,j-1) + 1,\ x_i = y_j \\ LCSS(i-1,j) \\ LCSS(i,j-1) \end{cases}$$

- **Mixed-type similarity:** for $X = (X_n, X_c)$ numeric/categorical parts,

$$S(X,Y) = \lambda S_{num}(X_n, Y_n) + (1-\lambda) S_{cat}(X_c, Y_c).$$

### 3.6 Information-theoretic measures

- **Entropy:**

$$H(X) = -\sum_{i=1}^{n} p_i \log_2 p_i, \qquad H \in [0, \log_2 n].$$

- **Mutual information:**

$$I(X,Y) = H(X) + H(Y) - H(X,Y).$$

## 4 Data Preparation (Priprema podataka)

- **Why:** quality of results depends on quality of data; raw data can be inconsistent, missing, noisy, heterogeneous.
- **Feature extraction (Izdvajanje karakteristika):** convert raw signals to informative features.
- **Type conversion (Prenosivost tipova):** transform representations to match algorithm assumptions; may lose information.
- **Discretization (Diskretizacija):** continuous $\to$ categorical by choosing number of bins $n$ and mapping values into sub-intervals.
- **Binarization (Binarizacija):** transform categorical values into asymmetric binary indicators (one-hot); beware high cardinality.
- **Text:** sparse term vectors are problematic for many methods; **LSA** can produce a lower-dimensional dense representation.
- **Vector normalization example:**

$$X \mapsto \frac{X}{\sqrt{\sum_{i=1}^{d} x_i^2}}.$$

- **Time series:**
  - **SAX:** windowed averaging + discretization using quantiles of a (usually assumed) normal distribution; often 3–10 intervals.
  - **DWT:** discrete wavelet transform (also DFT can be used).

# 5 Visualization (Vizuelizacija)

Common techniques to spot patterns and outliers:
- histograms, box-plots
- scatter plots (2D/3D)
- contour/heat maps
- parallel coordinates; star plots
- Chernoff faces
- Voronoi diagrams
- graphs/networks; tables; animations

# 6 Classification (Klasifikacija)

## 6.1 Problem setup

- Learn a model mapping attributes $X$ to a target $y$ (classification if $y$ is categorical; regression if numeric).
- Typical split: **training set** (build model) + **test set** (estimate generalization).

## 6.2 Decision trees (Drveta odlučivanja)

### 6.2.1 General induction procedure

Let $D_t$ be training records in node $t$, and classes $\{y_1, \ldots, y_c\}$.
- If $D_t$ is *pure* (single class), make a leaf.
- Else choose a test attribute (greedy) that optimizes an impurity-based split criterion; recurse on children.

### 6.2.2 Impurity measures and split criteria

Let $p(i \mid t)$ be relative class frequency in node $t$.
- **Gini:**

$$Gini(t) = 1 - \sum_{i=1}^{c} p(i \mid t)^2$$

- **Entropy:**

$$Entropy(t) = -\sum_{i=1}^{c} p(i \mid t) \log_2 p(i \mid t) \qquad (0 \cdot \log_2 0 := 0)$$

- **Classification error:**

$$CE(t) = 1 - \max_i p(i \mid t)$$

- **Gain** for a split into $k$ children $v_j$ with sizes $N(v_j)$, parent size $N$:

$$\Delta = Impurity(parent) - \sum_{j=1}^{k} \frac{N(v_j)}{N} Impurity(v_j).$$

- **Split information & gain ratio:**

$$split_info = -\sum_{j=1}^{k} P(v_j) \log_2 P(v_j), \qquad gain_ratio = \frac{\Delta}{split_info}.$$

### 6.2.3 Attribute tests (practical notes)

- Binary: always two branches.
- Nominal: multi-way split or binary partition into two groups.
- Ordinal: often prefer binary splits to preserve ordering.
- Continuous: choose threshold $A < \theta$ vs $A \geq \theta$ (or multi-interval); typically requires sorting.

### 6.2.4 Overfitting, pruning, and MDL

- **Underfitting:** model too simple (high train & test error).
- **Overfitting:** model too complex; low train error but worse test error.
- **Pruning styles:**
  - pre-pruning (early stopping)
  - post-pruning (grow fully then cut): reduced-error pruning; pessimistic error estimates; confidence-interval pruning
- **MDL principle:** choose the model minimizing

$$Cost(model, data) = Cost(model) + Cost(data \mid model).$$

### 6.2.5 Missing values (NV) handling (typical strategies)

- Adjust gain/gain-ratio for attributes with missing values.
- Distribute training instances with NV across children proportionally.
- For test-time NV: route probabilistically through branches and aggregate class probabilities.

### 6.2.6 Evaluation metrics

- **Confusion matrix** (binary): TP, TN, FP, FN.
- **Accuracy:** $\dfrac{TP + TN}{TP + TN + FP + FN}$ (can be misleading with class imbalance).
- **Precision / Recall:**

$$p = \frac{TP}{TP + FP}, \qquad r = \frac{TP}{TP + FN}$$

- **$F$-measure and $F_\beta$:**

$$F = \frac{2pr}{p + r}, \qquad F_\beta = \frac{(1 + \beta^2)pr}{\beta^2 p + r}$$

## 6.3 Decision tree algorithms (Algoritmi drveta odlučivanja)

### 6.3.1 ID3

- Uses **entropy** + **information gain**.
- Typically multi-way splits; no numeric-attribute handling (in the basic form), no missing values, no pruning; can overfit.

### 6.3.2 C4.5 (and J4.8, C5.0 family)

- Extends ID3: handles nominal + numeric attributes (numeric via threshold $\theta$).
- Uses **gain ratio** to reduce bias toward many-valued attributes.
- Post-pruning: reduced-error, pessimistic, and confidence-interval-based methods; also rule-based pruning via rules extracted from the tree.
- C5.0: fewer rules with similar accuracy; often combined with boosting.

### 6.3.3 CART (Classification and Regression Trees)

- Binary recursive partitioning; supports classification and regression.
- Grow large then prune via **cost-complexity pruning**; select best tree by test-set performance.
- Splits: numeric $X \leq C$; categorical via membership in a value subset.

### 6.3.4 CHAID, QUEST, SLIQ, SPRINT

- **CHAID:** chi-square tests for splitting.
- **QUEST:** fast/unbiased statistical trees.
- **SLIQ/SPRINT:** scalable variants; often use sorted/vertical representations.

## 6.4 Additional classification methods (Dodatne metode klasifikacije)

### 6.4.1 Rule-based classifiers (Klasifikatori zasnovani na pravilima)

- Learn if–then rules; common quality estimates include precision-like measures (and variants such as Laplace/M-estimate).

### 6.4.2 Instance-based (Klasifikatori zasnovani na instancama)

- **$k$-NN:** assign the majority class among $k$ nearest neighbors; sensitive to distance choice and feature scaling.

### 6.4.3 Artificial neural networks (Veštačke neuronske mreže)

- **Perceptron:** linear separator.
- **MLP + backprop:** minimize squared error

$$E(w) = \frac{1}{2} \sum_i (y_i - \hat{y}_i)^2.$$

### 6.4.4 Support Vector Machine (SVM)

- Maximum-margin hyperplane $w \cdot x + b = 0$.
- Hard-margin primal:

$$\min \frac{\|w\|^2}{2} \quad \text{s.t.} \quad y_i(w \cdot x_i + b) \geq 1.$$

- Dual (one common form):

$$L_D = \sum_i \lambda_i - \frac{1}{2} \sum_i \sum_j \lambda_i \lambda_j y_i y_j \, (x_i \cdot x_j).$$

- Soft margin uses slack variables $\xi_i$ and penalty $C$.
- Kernels: linear, polynomial, RBF $\exp(-\gamma \|x - y\|^2)$.

### 6.4.5 Ensembles (Klasifikacija ansamblom metoda)

- **Bagging**, **boosting**, **random forests** (variance reduction / accuracy improvements).

# 7 Cluster analysis (Klaster analiza)

## 7.1 Key cluster notions

- Exclusive vs non-exclusive; crisp vs fuzzy; partial vs complete; homogeneous vs heterogeneous.
- Cluster definitions: center/prototype-based (centroid/medoid), graph-based, neighborhood-based, density-based, conceptual clusters.

## 7.2 Representative-based algorithms (Algoritmi zasnovani na reprezentativnim predstavnicima)

Given points $X_1, \ldots, X_n$, find representatives $Y_1, \ldots, Y_k$ minimizing:

$$O = \sum_{i=1}^{n} \min_{1 \le j \le k} Dist(X_i, Y_j).$$

### 7.2.1 $k$-means (Algoritam k-sredina)

- Alternates assignment to nearest centroid and centroid recomputation.
- Initialization matters (random; **$k$-means++** spreads initial centroids).
- Complexity (typical): $O(n\, k\, \ell\, d)$.
- Objective in Euclidean space: **SSE**

$$SSE = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - c_i\|^2.$$

- Practical issues: local optima; outliers; empty clusters (replace by a point that reduces SSE).
- **Bisecting $k$-means:** repeatedly split clusters until $k$ clusters (e.g., split largest or highest-SSE cluster).

### 7.2.2 $k$-median (Algoritam k-medijana)

- Prototype is per-attribute median; typically uses Manhattan distance; less sensitive to outliers.

### 7.2.3 $k$-medoids (Algoritam k-medoida)

- Prototypes are actual data points (good for complex objects, e.g., time series); swap medoid candidate if it reduces $O$.

## 7.3 Hierarchical clustering (Hijerarhijsko klasterovanje)

### 7.3.1 Agglomerative (Sakupljajuće) clustering

- Start with $n$ singleton clusters; repeatedly merge closest clusters using a distance matrix ($O(n^2)$ memory).
- Linkages: single (min), complete (max), average, centroid, Ward.
- Ward-style SSE objective:

$$E = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - c_k\|^2.$$

- Lance–Williams update (generic):

$$p(R, Q) = \alpha_A p(A, Q) + \alpha_B p(B, Q) + \beta p(A, B) + \gamma \left| p(A, Q) - p(B, Q) \right|.$$

### 7.3.2 Divisive (Razdvajajuće) clustering

- Start with one cluster; repeatedly split (e.g., bisecting $k$-means).

## 7.4 Grid/density-based clustering (Algoritmi zasnovani na mrežama i gustini)

### 7.4.1 DBSCAN

Parameters: radius $\varepsilon$ and min-neighbors $t$.
- **Core:** at least $t$ neighbors within $\varepsilon$.
- **Border:** fewer than $t$ neighbors but within $\varepsilon$ of a core point.
- **Noise:** neither core nor border.
- Finds arbitrary-shaped clusters; struggles with varying density.

## 7.5 Cluster validation (Provera korektnosti klasterovanja)

- Clustering always returns some partition even if no natural clusters exist; compare multiple algorithms/parameters.
- **Hopkins statistic:**
$$H = \frac{\sum_{i=1}^{p} v_i}{\sum_{i=1}^{p} o_i + \sum_{i=1}^{p} v_i}$$
($H \approx 0.5$ random; $H \to 1$ clusterable).
- **Silhouette** for point $i$:
$$S_i = \frac{b_i - a_i}{\max\{a_i, b_i\}}, \quad a_i = AvgDist_{in}^{i},\ b_i = MinDist_{out}^{i}.$$

## 7.6 Clustering categorical data (Klasterovanje kategoričkih podataka)

- Avoid naive one-hot when cardinality is huge.
- **$k$-modes:** prototype uses per-attribute mode; can normalize frequencies when distributions are skewed.
- **$k$-medoids:** prototype is a real record.

## 7.7 Subspace clustering (Klasterovanje po potprostorima)

- Clusters may exist only in subsets of attributes (different clusters in different subspaces).
- Algorithms:
  - **CLIQUE:** grid + density; Apriori-like search for dense subspaces.
  - **MAFIA:** CLIQUE variant selecting low-entropy subspaces (can detect low-density clusters; expensive).
  - **ENCLUS:** entropy-based subspace selection; histogram-based binning; parallelizable.
  - **DENCLUE:** influence functions; density is sum of influences; clusters around attraction points (local maxima).

## 7.8 Scalable clustering (Klasterovanje skalabilnih podataka)

- **CLARA/CLARANS:** scalable medoid-based methods.
- **CURE:** hierarchical using representatives.
- **BIRCH:** incremental hierarchical clustering features.

### 7.9 Self-organizing maps (Samoorganizujuće mape, SOM)

- 2D topographic grid of centroids; update winner and its neighbors.
- Update rule for winner centroid $m_j$ at time $t$:

$$m_j(t+1) = m_j(t) + h_j(t)\,(p(t) - m_j(t)).$$

- Pros: preserves neighborhood relations; good for visualization/compression. Cons: parameter tuning; no explicit global objective; no strict convergence guarantee.

## 8 Anomaly detection (Otkrivanje anomalija)

### 8.1 Introduction (Uvod)

- **Anomaly/outlier:** observation so different that it suggests a different generating mechanism.
- Distinguish **noise** (random error, not interesting) vs **anomaly** (interesting if not caused by noise).
- High dimensionality makes anomalies harder; anomalies can be global or local (context-dependent).
- Often use a **score** and a threshold, or take top-$n$ anomalies.

### 8.2 Main approaches

- **Model-based:** learn "normal" model; anomalies violate it (unsupervised: rare clusters).
- **Statistical:** anomalies have low probability under assumed distribution; for normal distribution use $[\mu - 2\sigma, \mu + 2\sigma]$ (95.4%) or $3\sigma$ ($> 99\%$); small $n$ may use Grubbs/Student-$t$.
- **Visualization-based:** manual detection from plots (few attributes).
- **Distance-based:** e.g., $k$-NN distance/average distance; sensitive to $k$ and expensive.
- **Density-based:** inverse neighbor distance; relative density; **LOF** compares local density to neighbors (large LOF $\Rightarrow$ outlier).
- **Clustering-based:** points far from prototypes / low-density points / weakly connected nodes.

## 9 Association rules (Pravila pridruživanja)

### 9.1 Basics and metrics

- Transaction $t$: ID + set of items. Rule: **body** $\rightarrow$ **head**.
- Itemset $A$: set of items; $k$-itemset has $k$ items.
- **Support count:** $\#(A)$ is number of transactions containing $A$.
- **Frequent itemset:** $\#(A) \geq minsup$.
- **Support of rule:**

$$sup(A \rightarrow B) = \frac{\#(A \cup B)}{N}$$

- **Confidence of rule:**

$$conf(A \rightarrow B) = \frac{\#(A \cup B)}{\#(A)}$$

- Goal: all rules with $sup \geq minsup$ and $conf \geq minconf$.

### 9.2 Frequent itemset mining (Algoritmi za određivanje čestog skupa stavki)

#### 9.2.1 Brute force (Metoda grube sile)

- Checks all candidates ($2^d - 1$ itemsets for $d$ items); impractical for large $d$.

### 9.2.2 Apriori

- **Apriori principle / antimonotonicity:**

$$X \subset Y \Rightarrow sup(Y) \leq sup(X).$$

  If an itemset is not frequent, no superset can be frequent.
- Level-wise candidate generation + pruning; often uses a hash-tree to reduce comparisons.
- Frequent itemset $I$ can generate up to $2^{|I|} - 2$ rules (non-empty LHS and RHS).
- **Maximal frequent itemsets:** no frequent immediate superset.
- **Closed itemsets:** no immediate superset with equal support.

### 9.2.3 FP-growth (Algoritam FP rasta)

- Builds an FP-tree by sorting items by decreasing frequency; compresses shared prefixes.
- Mines frequent patterns bottom-up via conditional FP-trees; avoids candidate explosion.

## 9.3 Interestingness measures (Mere interesantnosti pravila)

Support/confidence can be misleading if RHS is very frequent.

- **Lift:**

$$Lift(A, B) = \frac{conf(A \to B)}{sup(B)} = \frac{sup(A, B)}{sup(A)sup(B)}$$

- **Leverage / Piatetsky–Shapiro:**

$$PS = sup(A, B) - sup(A)sup(B)$$

- **Interest ratio (odnos kamate) for items $i_1, \ldots, i_k$:**

$$I(i_1, \ldots, i_k) = \frac{sup(i_1, \ldots, i_k)}{\prod_{j=1}^{k} sup(i_j)}$$

- **Deployability:**

$$D = \frac{sup(A) - sup(A, B)}{N} \quad \text{(closer to 0 is better)}$$

- **Pearson correlation (for contingency table counts $f$):**

$$\rho = \frac{f_{11}f_{00} - f_{01}f_{10}}{\sqrt{f_{1+}f_{+1}f_{0+}f_{+0}}}$$

- **Chi-square:**

$$\chi^2(X) = \sum_{i=1}^{2^{|X|}} \frac{(O_i - E_i)^2}{E_i}$$

- **IS measure (asymmetric binaries):**

$$IS(A, B) = \frac{sup(A, B)}{\sqrt{sup(A)sup(B)}} \quad \equiv \quad \cos(A, B)$$

- **All-confidence (for itemset $X$):**

$$all_confidence(X) = \frac{sup(X)}{\max_{x \in X} sup(x)}$$

- **Simpson's paradox:** correlations can flip when aggregating subgroups.

## 9.4 Additional association-rule techniques (Dodatne tehnike pravila pridruživanja)

### 9.4.1 Categorical attributes (Kategorički atributi)

- Convert to asymmetric binary indicators; challenges: high-cardinality values (low support).
- Mitigations: value aggregation; drop extremely frequent items that dominate.

### 9.4.2 Continuous attributes (Neprekidni atributi)

- **Discretization-based:** unsupervised (equal-width/equal-depth/binning or clustering) or supervised (separate classes well); interval size affects support vs confidence.
- **Statistical rules:** RHS is a statistic (mean/median, etc.); compare covered vs uncovered population via hypothesis testing (e.g., $Z$-test).
- **No-discretization:** discover relations between continuous vectors after normalization (e.g., word co-occurrence in TF matrices with per-word support normalized to 1).

### 9.4.3 Multi-level rules (Pravila pridruživanja između više nivoa)

- Use a concept hierarchy so low-level patterns can be supported by higher-level generalizations.
- Key monotonicity-style implications from the script (idea-level): parents have higher support; if a detailed pattern is frequent, corresponding generalized patterns are also frequent.

### 9.4.4 Sequential data (Niz podataka) and sequential patterns

- Sequence $s = \langle e_1 e_2 \dots \rangle$; element $e_i$ contains an itemset.
- Subsequence relation: $a$ is contained in $b$ if items match in order across increasing indices.
- **Support of a sequence:** fraction of sequences containing it.
- **GSP algorithm:** level-wise candidate join + pruning using frequent $(k-1)$-subsequences.
- Time constraints: max-span, min-gap, max-gap; window size; continuous subsequences help preserve Apriori-style pruning when max-gap breaks it.

### 9.4.5 Rare and negative patterns (Retki i negativni obrasci)

- **Rare pattern:** support below *minsup* (can still be interesting).
- Negative correlation (for itemset $X = \{x_1, \dots, x_k\}$):

$$sup(X) < \prod_{j=1}^{k} sup(x_j).$$

- Negative rule $X \to Y$ (disjoint $X, Y$):

$$sup(X \cup Y) < sup(X) sup(Y).$$

- Mining negatives is hard: item universe explodes; support pruning is less effective.

### 9.4.6 Associative classification (Klasifikacija pomoću pravila pridruživanja)

- Use high-confidence rules where the RHS is a single class label; LHS contains attribute-value items.